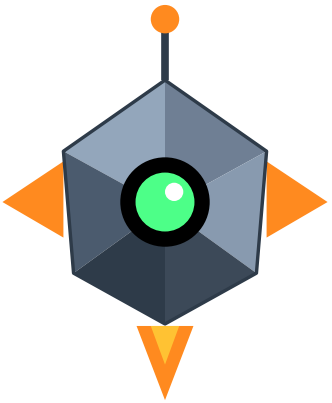
BOOK 4

The Order of Discovery

From messy notebooks to tidy proofs





MULTIX // HELLO
I am Multix. I ride along in every chapter. Watch my readout: it shows what is owed, what is paid, and what is only a guess.

CHAPTER 1

Messy Notebook, Tidy Proof



MULTIX // CIRCLE CHECK
A circle holds a piece that belongs together, and circles can sit inside circles. A circle is *ready* when everything needed for its calculation is known and its operation is allowed by the current rules. To **pop** a ready circle, replace it with its value and leave everything outside it alone. To **redraw**, move circles around or split a number, as long as the value stays the same. The full rules are in the introduction.

STORY

Kofi had a puzzle from his teacher:
Find two numbers that add up to 20 and multiply to 96.
He sat down with a pencil and a fresh page, and he did what mathematicians actually do. He made a mess.

STEP 1 Kofi's notebook

Here is his page, exactly as he wrote it, crossed-out parts included.

- Try 10 and 10. They add to 20. Multiply: 100. **Too big.**
- Try 5 and 15. Add to 20. Multiply: 75. **Too small. Jumped too far.**
- ~~Try 6 and 16. Multiply: 96!!~~ Wait. 6 plus 16 is 22, not 20. **Broken rule. Cross it out.** (It did multiply to 96, but the sum is wrong, so it does not count.)
- Try 9 and 11. Add to 20. Multiply: 99. **Close!**
- Try 8 and 12. Add to 20. Multiply: 96. **Yes!**

Kofi had his answer: 8 and 12. He also had a page full of guesses, a crossed-out mistake, and a feeling that he had been wandering.

But look again at the guesses that kept the sum at 20. Kofi made a small list:

Pair	Product	Missing from 100
10 and 10	100	0
9 and 11	99	1

Pair	Product	Missing from 100
8 and 12	96	4
7 and 13	91	9

The last column goes 0, 1, 4, 9. Those are square numbers. Every step away from 10 takes away a square.

Kofi had not been wandering at all. His guesses were quietly teaching him a pattern.

IN CIRCLES

Each of Kofi's guesses is a circle he draws and pops, and then compares with the target:

Trust check. Kofi's messy notebook gave him an answer.

TRACE

```
( 10 × 10 )  →  100    too big
( 9 × 11 )   →  99     closer
( 8 × 12 )   →  96     yes
( 8 + 12 )   →  20     the sum matches
```

His messy notebook is a long trace of pops, including the ones he crossed out.

STEP 2 The tidy proof

Kofi wanted a shorter version that made the reasoning easy to follow. So he wrote a second version:

> Let *d* be the distance from 10, so *d* is 0 or more. Let the two numbers be 10 minus *d* and 10 plus *d*. They add up to 20.
>
> Their product is 100 minus *d* times *d*.
>
> We want the product to be 96, so *d* times *d* is 4, which means *d* is 2.
>
> The numbers are 8 and 12. Check: 8 + 12 = 20 and 8 × 12 = 96.

That is the whole proof. It has no guesses and no crossed-out parts, every line follows from the one before it, and the last line checks the answer in the original puzzle.

IN CIRCLES

The tidy proof is a trace too, but a much shorter one. It starts with a redraw:

TRACE

```
( ( 10 − d ) × ( 10 + d ) )
redraw  →  ( ( 10 × 10 ) − ( d × d ) )
```

That is the missing corner from The Work That Vanishes: a 10 by 10 square with a *d* by *d* corner taken away. The tidy trace keeps only the redraw and the pops that matter. The messy notebook keeps every pop, wrong turns included.

STEP 3 Why have two versions?

The two pages do different jobs.

The **messy notebook** is where the idea was *found*. Without the guesses, Kofi would not have noticed the squares. The wrong turns were not wasted. They were how he learned what to look for.

The **tidy proof** is where the idea is *checked*. Anyone can read it, line by line, and decide for themselves whether it is true. They do not need to follow Kofi's wandering.

Each version leaves something out. The tidy proof hides how Kofi found the answer, which can make it look like magic: "How did he know to write 10 minus *d*?" The messy notebook hides whether the answer is correct, because you have to read every attempt to be sure nothing went wrong.

A good mathematician keeps both. The notebook is for finding. The proof is for sharing. (A teacher is often glad to see the notebook too, because it shows how you think.)

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the tidy proof keep from the messy page?** (The answer, and a reason why it works.)
- **What did it throw away?** (The wrong guesses, the crossed-out mistake, and the order in which things were tried.)
- **Was anything thrown away that mattered?** (Yes. The guesses showed Kofi the pattern of squares. The proof does not say where the idea of 10 minus *d* came from.)

WHAT MAKES THIS ANSWER TRUSTWORTHY?

Kofi's messy notebook gave him an answer. What made it trustworthy was a check: 8 plus 12 is 20, and 8 times 12 is 96. The notebook found the answer, and the check confirmed it.

The tidy proof is trustworthy in a different way. It gives a reason why the answer must be right, because every line follows from the one before. It still ends with a check, because a check can catch a slip in the reasoning.

(Notice something else: the 10 minus d and 10 plus d trick is the missing-corner trick from The Work That Vanishes. 10 times 10 minus a corner is exactly what the tidy proof used.)

PRACTICE

1. Find two numbers that add to 30 and multiply to 221. Keep a messy notebook first. Then write a tidy version in three or four lines.
2. Find two numbers that add to 20 and multiply to 99. Use Kofi's tidy method without any guessing. What is d this time?
3. Find two numbers that add to 20 and multiply to 91.
4. **The impossible puzzle.** Try to find two numbers that add to 20 and multiply to 101. Keep a messy notebook. What do you notice about your products? Then use the tidy method. What goes wrong with it?
5. Kofi tries 7 and 13. Write the product as a circle and pop it. Does the pair work?

BRIDGE *Bridge to the rest of the series:* The messy notebook is how people really find things. The tidy proof is how they show things. Both books before this one used the tidy version. In A Useful Wrong Answer, the guess about 7/8 was a messy-notebook move. In Two Boxes, One Answer, trying numbers from 1 to 9 was a messy-notebook move, and finding the reason that A and B always match was the tidy proof.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **1,225**. Leave the others blank. What appears?

1217 + 18	1196 + 29	1240 − 15	1207 + 20	1222 + 12	30 × 40 + 5 × 5	1236 − 11	1216 + 20
1223 + 2	1206 + 19	1222 + 3	30 × 40 + 5 × 5	175 × 7	1242 − 17	1206 + 19	1199 + 26
22 × 48 + 13 × 13	18 × 52 + 17 × 17	16 × 54 + 19 × 19	1193 + 32	26 × 44 + 9 × 9	1216 + 9	18 × 52 + 17 × 17	1197 + 28
1252 − 27	21 × 49 + 14 × 14	11 × 59 + 24 × 24	35 × 35	17 × 53 + 18 × 18	1249 − 24	25 × 49	91 × 91 − 84 × 84
1194 + 28	1221 + 4	11 × 59 + 24 × 24	175 × 7	1196 + 29	1203 + 22	1219 + 6	1218 + 9
1205 + 12	1219 + 11	1237 − 12	1204 + 21	1245 − 20	1231 − 6	1212 + 12	1231 + 3
1206 + 29	1187 + 28	1212 + 20	1202 + 23	1261 − 36	1211 + 21	1201 + 18	1226 + 3
1197 + 16	6 × 205	1215 + 20	1205 + 8	1209 + 4	1207 + 11	1208 + 27	1208 + 15



MULTIX//SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 2

The Pattern That Held, Until It Didn't

STORY

After Kofi cleaned up his notebook in chapter 1, he started collecting patterns. A pattern is exciting, because it feels like a discovery. But Kofi's teacher gave him two patterns on the same day, and they turned out very differently.

PATTERN 1: ADDING ODD NUMBERS

Kofi added the odd numbers from the start, one more each time, and wrote each total in his notebook.

TRACE

```
( 1 + 3 )              →  4
( 1 + 3 + 5 )          →  9
( 1 + 3 + 5 + 7 )      →  16
( 1 + 3 + 5 + 7 + 9 )  →  25
```

"4, 9, 16, 25," said Kofi. "Those are 2 × 2, 3 × 3, 4 × 4 and 5 × 5. The totals are squares!"

He wondered if it would keep going. The next total should be 6 × 6 = 36. He added the next odd number, 11, to 25 and got 36. It held again.

PATTERN 2: A RECIPE FOR PRIME NUMBERS

A prime number is a number bigger than 1 that can only be shared out evenly by 1 and by itself. Examples are 2, 3, 5, 7, 11, 13. The number 15 is not prime, because 3 × 5 = 15.

Kofi's teacher gave him a recipe: *take a number n, multiply it by itself, add n, then add 41.* In circles:

TRACE

```
( ( n × n ) + n ) + 41
```

Kofi tried the first five:

Trust check. A pattern is trustworthy when you can explain *why* it must continue, as the L-border explains the odd numbers.

n	((n × n) + n) + 41	Prime?
1	43	yes
2	47	yes
3	53	yes
4	61	yes
5	71	yes

“Five out of five!” said Kofi. “The recipe makes primes.”

The teacher smiled. “Try n = 40.”

TRACE

```
( ( 40 × 40 ) + 40 ) + 41
pop ( 40 × 40 )  →  1,600
( ( 1,600 + 40 ) + 41 )
pop ( 1,600 + 40 )  →  1,640
( 1,640 + 41 )
pop  →  1,681
```

And 1,681 is not prime. It is 41 × 41. The recipe worked for 40 numbers in a row, from n = 0 to n = 39, and then it broke.

STEP 1 Same kind of evidence, very different outcomes

Both patterns looked the same in Kofi’s notebook. Each had a handful of examples that fit. Yet one of them is always true and the other is not.

So what is different? Not the number of examples. The recipe for primes had *forty* working examples, which is a lot more than the five that Kofi tried for the odd numbers. What is different is that one pattern comes with a **reason**.

STEP 2 The reason for the odd numbers

Picture a square of dots, 4 by 4, that is 16 dots. To make the next square, 5 by 5, you add a border along two sides, shaped like an L. How many dots are in the border? 4 along one side, 4 along the other, and 1 in the corner. That is 4 + 4 + 1 = **9**, which is the next odd number.

In circles:

TRACE

```
( 4 × 4 )  →  16
add the L:  ( 16 + 9 )  →  25  =  ( 5 × 5 )
```

This works for any square, not just 4 by 4. For a square of side *n*, the L-border has *n* dots along one side, *n* along the other, and 1 in the corner, which is *n* + *n* + 1. That is always the next

odd number. So every time you add the next odd number, you get the next square.

That is a reason. It does not depend on trying more examples. It says why the pattern must continue forever.

STEP 3 Is there a reason for the prime recipe?

The recipe has no L-border or anything like it. Nobody found a reason that makes it always give primes, and there is none, because at n = 40 it fails. The forty working examples were a very lucky run.

When a pattern fails, it often fails somewhere you have not looked yet. There is no way to know ahead of time how far to look. That is why a pattern, however long it holds, is called a **conjecture** until it has a reason.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the examples keep for both patterns?** (A list of cases that fit.)
- **What did the examples not keep?** (Any promise about cases nobody has tried.)

WHAT MAKES THIS TRUSTWORTHY?

A pattern is trustworthy when you can explain *why* it must continue, as the L-border explains the odd numbers. Examples can make you suspect a pattern, and they can show it to be false. But they cannot show it true for every case, if there are endlessly many cases.
When Kofi writes a pattern in his notebook, he now marks it as one of two things: **has a reason** or **only seen so far**.

PRACTICE

1. Add the first six odd numbers, 1 + 3 + 5 + 7 + 9 + 11. What square do you get?
2. Use the L-border idea to explain why adding 13 to 36 gives the next square. What is it?
3. Try the recipe ((n × n) + n) + 41 for n = 6. Is the answer prime? (Hint: it is 83.)
4. Dani says, “I tested 100 numbers and the pattern held every time, so it must be true forever.” What would you say to Dani?

5. At n = 41, the recipe gives ((41 × 41) + 41) + 41. Show that the answer can be divided by 41.

6. Sort these into “has a reason” and “only seen so far”: (a) the odd numbers add up to squares; (b) every number I tried in the prime recipe gave a prime.

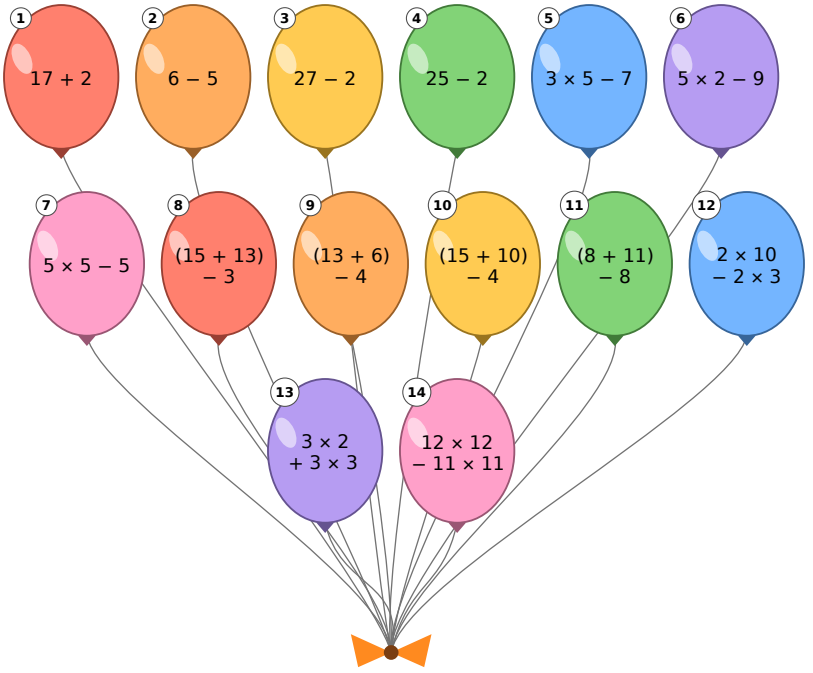
BRIDGE *Bridge to the rest of the series:* In chapter 1, Kofi’s tidy proof ended with a check. Here the check has a new job: telling the difference between a pattern with a reason and a pattern that is only seen so far. In Two Boxes, One Answer, the detective’s rule says the same thing about boxes: finite tests cannot prove two things equal in every case.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

													!
1	2	3	4	5	6	7	8	9	10	11	12	13	14

CHAPTER 3

The Trick of the Pairs

STORY

Kofi's teacher gave the class a long job: *add up every number from 1 to 10.* Kofi wrote down what he did, mess and all.

STEP 1 The messy notebook

TRACE

```
1 + 2 = 3
3 + 3 = 6
6 + 4 = 10
10 + 5 = 15
15 + 6 = 21
21 + 7 = 28
28 + 8 = 36
36 + 9 = 45
45 + 10 = 55
```

That took nine lines. While writing the lines, Kofi noticed something odd. The first number and the last number added to 11. So did the second and the second-to-last.

TRACE

```
1 + 10 = 11
2 + 9  = 11
3 + 8  = 11
4 + 7  = 11
5 + 6  = 11
```

Five pairs, and every pair makes 11. "So I don't need nine lines," Kofi said. "I need five elevens."

STEP 2 The tidy notebook

Kofi rewrote the whole thing, leaving out the dead ends and the order of his discovery.

Trust check. The reason is that pairing the ends always gives the same sum.

TRACE

```
( 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 )
redraw  →  ( ( 1 + 10 ) + ( 2 + 9 ) + ( 3 + 8 ) + ( 4 + 7 ) + ( 5 + 6 ) )
pop all five  →  ( 11 + 11 + 11 + 11 + 11 )
redraw  →  ( 5 × 11 )
pop  →  55
```

The redraw keeps the value, because adding the same numbers in a different order and grouping gives the same total. The answer matches the messy notebook: 55.

STEP 3 Does it work for a bigger job?

The tidy notebook does something the messy one cannot: it tells Kofi what to do for 1 to 100 without writing a hundred lines.

There are 100 numbers, so 50 pairs. The first and last make 101, and so do the second and second-to-last.

TRACE

```
( 50 × 101 )  →  5,050
```

The sum of 1 to 100 is **5,050**.

What about an odd number of numbers, such as 1 to 9? Now the pairing leaves one number alone in the middle.

TRACE

```
( 1 + 9 ) ( 2 + 8 ) ( 3 + 7 ) ( 4 + 6 )  and 5 left over
( 4 × 10 ) + 5  →  45
```

So the sum of 1 to 9 is **45**.

STEP 4 Why keep the messy notebook?

Kofi nearly threw the messy notebook away. But look at what it had that the tidy one did not. It had the **nine-line check**: 55 from adding one at a time, a result that came without the pairing idea. When the pairing gave 55 too, Kofi trusted both more.

It also had the **discovery**. Nobody would have invented the pairing without noticing the pattern of 11s while writing out the slow way.

A good habit: keep the messy one for finding the idea and for checking, and use the tidy one for telling other people.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the tidy notebook keep from the messy one?** (The answer, 55, and the reason it is right.)
- **What did it drop?** (The order of discovery, and the nine slow lines.)

WHAT MAKES THIS TRUSTWORTHY?

The reason is that pairing the ends always gives the same sum. When you move one step in from the left, the number goes up by 1. When you move one step in from the right, the number goes down by 1. So each pair keeps the same total. That works for any length, not only 10 or 100.
The messy notebook is a second, separate check: it gives the same answer by a different road.

PRACTICE

1. Add 1 to 20 using pairs. How many pairs? What does each pair make?
2. Add 1 to 8 using pairs.
3. Add 5 to 14 (that is 5, 6, 7, ..., 14). Be careful: what does the first pair make? How many pairs are there?
4. Add 1 to 99. This is an odd count, so one number is left in the middle. Which one?
5. Kofi says “the tidy notebook is all we need, so we can throw the messy one away.” What would you say?

BRIDGE *Bridge to the rest of the series:* In chapter 1, the tidy proof ended with a check. Here the messy notebook is the check, and the pairing idea is the discovery. In Reversing the Calculation, a diary of each press is what lets you go back, and a messy notebook is a diary of your thinking.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 38 × 42 )
2 redraw  →  ( ( 40 − 2 ) × ( 40 + 2 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 2 × ( 2 + 1 ) ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 2 × ( 2 + 1 ) )  →  6
6 pop ( 1,600 − 6 )  →  1,594
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 24 × 26 )
2 redraw  →  ( ( 25 − 1 ) × ( 25 + 1 ) )
3 redraw  →  ( ( 25 × 25 ) − ( 1 × 1 ) )
4 pop ( 25 × 25 )  →  625
5 pop ( 1 × 1 )  →  2
6 pop ( 625 − 2 )  →  623
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 46 × 54 )
2 redraw  →  ( ( 50 − 4 ) × ( 50 + 4 ) )
3 redraw  →  ( ( 50 × 50 ) − ( 4 × 4 ) )
4 pop ( 50 × 50 )  →  2,500
5 pop ( 4 × 4 )  →  16
6 pop ( 2,500 − 16 )  →  2,484
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 4

The Question Nobody Has Settled

STORY

Kofi found a game with two rules.

- If your number is **even**, cut it in half.
- If your number is **odd**, multiply it by 3 and add 1.

Then do the same again to the new number, and keep going. Kofi's job was to see where the numbers end up.

STEP 1 Try 6

TRACE

```
6    even   →  ( 6 ÷ 2 )  →  3
3    odd    →  ( ( 3 × 3 ) + 1 )  →  10
10   even   →  5
5    odd    →  ( ( 5 × 3 ) + 1 )  →  16
16   even   →  8
8    even   →  4
4    even   →  2
2    even   →  1
```

Kofi reached **1** after 8 moves.

STEP 2 Try 7

TRACE

```
7 → 22 → 11 → 34 → 17 → 52 → 26 → 13 → 40 → 20 → 10 → 5 → 16 → 8 → 4 → 2 → 1
```

This took 16 moves. The numbers went up as high as 52 before coming down. But they reached 1.

Trust check. The individual trails are trustworthy, because each move can be checked: halving an even number and tripling an odd one plus 1 are plain pops.

STEP 3 A pattern, but is it a promise?

Kofi tried 9, 12, and 27. Every one ended at 1. He wrote in his notebook:

Every positive whole number, used as a starting number, reaches 1.

Is that a promise? He looked for a reason, as in chapter 2. He could not find one.

Here is what people know. Computers have tried this game on a huge range of starting numbers, far beyond anything one could do by hand, and every positive whole number reached 1. But **nobody has found a reason that works for every positive whole starting number**, and nobody has found one that fails. It is a famous open question.

So Kofi marked his notebook:

Every positive whole number, used as a starting number, reaches 1. **Only seen so far.**

This is different from the odd-number pattern in chapter 2, which had a reason, and different from the prime recipe, which was false. This one is still open.

STEP 4 Why bother, then?

An open question teaches two things.

First, a pattern can survive an enormous number of tests and still have no reason behind it. Surviving tests is a kind of evidence, but evidence is not a reason.

Second, being honest about what is unsettled is part of good mathematics. Kofi's notebook can say *I do not know yet* and still be a good notebook.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the many examples keep?** (A list of numbers that reach 1.)
- **What is missing?** (A reason that covers all of them.)

WHAT MAKES THIS TRUSTWORTHY?

The individual trails are trustworthy, because each move can be checked: halving an even number and tripling an odd one plus 1 are plain pops. The claim "every positive whole number reaches 1" is not, and the right label is *only seen so far*.
A trustworthy notebook keeps these two things apart: what has been shown, and what has only been noticed.

PRACTICE

1. Follow 5 until it reaches 1. How many moves?
2. Follow 10. How many moves? Compare it with the trail for 5.

3. Follow 12 to 1. How many moves?

4. Follow 1 for three moves. What do you notice?

5. Kofi tried 20 numbers and all reached 1. Can he write “has a reason” in his notebook? What should he write?

BRIDGE *Bridge to the rest of the series:* In chapter 2, patterns were sorted into “has a reason” and “only seen so far”. This chapter gives a pattern that stays in the second box for everyone, not just for Kofi. In What Must Remain Unresolved, leaving a question open is part of the work, not a failure.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **899**. Leave the others blank. What appears?

5 × 181	911 − 12	29 × 31	887 + 11	55 × 55 − 46 × 46	936 − 37	869 + 30	878 + 27
874 + 25	910 − 11	869 + 30	31 × 29	921 − 22	868 + 31	928 − 29	29 × 31
879 + 20	934 − 35	880 + 19	881 + 18	875 + 24	924 − 25	873 + 26	895 + 4
912 − 13	891 + 8	931 − 32	862 + 37	892 + 7	903 − 4	929 − 30	935 − 36
871 + 17	866 + 33	863 + 36	891 + 8	869 + 30	914 − 15	916 − 17	904 + 4
883 + 15	891 + 14	877 + 22	930 − 31	909 − 10	871 + 28	886 + 23	909 + 2
871 + 23	898 + 12	876 + 28	860 + 39	926 − 27	878 + 14	879 + 23	895 + 10
874 + 16	896 + 7	870 + 18	882 + 29	871 + 20	884 + 3	873 + 20	883 + 24



MULTIX // SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 5

Why Odd Times Odd Is Odd

STORY

An odd number is one that leaves 1 over when you share it in twos. Kofi multiplied some odd numbers together.

TRACE

```
( 3 × 5 )    →  15
( 7 × 9 )    →  63
( 11 × 13 )  →  143
( 5 × 5 )    →  25
```

All four answers are odd. Kofi wrote a pattern in his notebook: *odd times odd is odd.*
He had a rule from chapter 2: before trusting a pattern, look for a reason. Is there one here?

STEP 1 A reason in one example

Take 7 × 9. Both are odd, which means each is an even number plus 1.

- 7 is 6 + 1.
- 9 is 8 + 1.

Kofi split both and multiplied the pieces.

TRACE

```
( 7 × 9 )
redraw  →  ( ( 6 + 1 ) × ( 8 + 1 ) )
redraw  →  ( ( 6 × 8 ) + ( 6 × 1 ) + ( 1 × 8 ) + ( 1 × 1 ) )
pop all four  →  ( 48 + 6 + 8 + 1 )
pop  →  63
```

Look at the four pieces: 48, 6, 8 and 1.

- 48 is even.
- 6 is even.
- 8 is even.
- 1 is the only odd one.

Three even numbers added together give an even number. Then adding 1 makes it odd. So the total, 63, must be odd.

STEP 2 The reason works for every pair

Trust check. The reason is the split into four pieces.

Nothing in that argument depended on 7 and 9. For any two odd numbers:

- the first is an even number plus 1,
- the second is an even number plus 1,
- multiplying gives four pieces: even times even, even times 1, 1 times even, and 1 times 1.

The first three pieces are even, and the last is 1. So the answer is even plus 1, which is odd. It works for 3 × 5, for 11 × 13, and for numbers with a hundred digits.

This is a reason, not just examples. Kofi can mark the pattern **has a reason**.

STEP 3 Contrast with a neighbour

What about odd plus odd? Try 7 + 9.

TRACE

```
( 7 + 9 )  →  16
```

Even. The reason: each odd number is an even number plus 1. The two leftover 1s make 2, which is even, so the whole sum is even.

What about even times odd? Try 6 × 7. That is six sevens. Six is even, so the sevens can be paired off with none left over.

TRACE

```
( 6 × 7 )  →  42
```

Even. The reason: if either number is even, the product has pairs with nothing left over.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the split keep?** (The value, 63, in four pieces.)
- **What did it show?** (Which pieces are even, and that only the 1 times 1 is odd.)

WHAT MAKES THIS TRUSTWORTHY?

The reason is the split into four pieces. It covers every pair of odd numbers, so it is stronger than any list of examples.
The examples are still useful. They suggested the pattern in the first place, and they check the arithmetic. As in chapter 3, the messy notebook finds the idea and the tidy one gives the reason.

PRACTICE

1. Split 15×21 into four pieces, add them, and say whether the answer is odd or even.
2. Is 13×17 odd or even? Explain without multiplying.
3. Is $14 + 9$ odd or even? Give a reason with the split idea.
4. Is 8×25 odd or even? Which rule decides it?
5. Kofi says: "Even plus even is even, because I tried $4 + 6$, $8 + 10$ and $2 + 12$." What reason could he give instead?

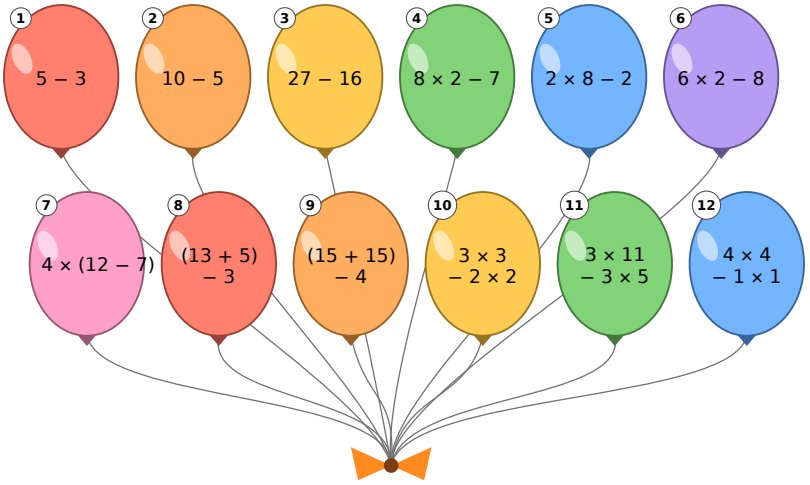
BRIDGE *Bridge to the rest of the series:* In chapter 2, examples alone could not tell a true pattern from a false one. Here a split gives a reason that covers every pair. In Two Boxes, One Answer, a reason is also what tells us two boxes really are the same.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

											!
1	2	3	4	5	6	7	8	9	10	11	12

CHAPTER 6

One Case Is Enough

STORY

Kofi wrote down a claim in his notebook:
Multiplying a number always makes it bigger.
He tried some examples.

TRACE

```
( 5 × 3 )  →  15     bigger than 5
( 5 × 2 )  →  10     bigger than 5
( 8 × 4 )  →  32     bigger than 8
```

Three examples, three times bigger. Is the claim true?

STEP 1 Look for a case that breaks it

Kofi's teacher said: "Try to break it. You only need one case."

Kofi thought about numbers he had not tried. What about multiplying by 1?

TRACE

```
( 5 × 1 )  →  5     the same as 5, not bigger
```

And by a number smaller than 1?

TRACE

```
( 5 × 0.5 )  →  2.5     smaller than 5
```

One case is enough. Half of 5 is 2.5, and 2.5 is smaller than 5. The claim is **false**. A case that breaks a claim is called a **counterexample**.

Kofi rewrote the claim so that it was true:

Multiplying a number by something bigger than 1 makes it bigger.

Now there is a reason: if you take more than one lot of a number, you have more than the number.

STEP 2 Counterexamples are strong

Compare what examples and counterexamples can do.

Trust check. A counterexample is trustworthy because anyone can check it by doing the arithmetic.

- A hundred examples that fit a claim do **not** prove it. There might be a case no one has tried.
- One counterexample **disproves** the claim, for certain.

This is a strange fact. A claim is easy to kill and hard to prove. That is why mathematicians love a good counterexample.

STEP 3 More claims to test

Claim: “Squaring a number always makes it bigger.”

TRACE

```
( 3 × 3 )  →  9     bigger
( 1 × 1 )  →  1     the same as 1
( 0.5 × 0.5 )  →  0.25     smaller than 0.5
```

False, with counterexamples.

Claim: “Dividing a number always makes it smaller.”

TRACE

```
( 6 ÷ 2 )  →  3     smaller
( 6 ÷ 0.5 )  →  12     bigger
```

False. Dividing by a number less than 1 makes it bigger.

Claim: “Adding a number to another whole number makes it bigger.” Is it true? Try adding 0: 5 + 0 = 5, not bigger. Add 1 or more, and the claim holds. The reason is that you are adding something.

Kofi sorted his notebook into three piles.

Pile	Meaning
has a reason	we can explain why it must be true
only seen so far	examples fit, no reason yet
refuted	a counterexample breaks it

STEP 4 How to look for a counterexample

Kofi learned where counterexamples like to hide.

- At the **edges**: 0, 1, and the very smallest or largest numbers.
- At the **special numbers**: one that is its own opposite, a number less than 1.
- At cases where the **claim says “always”** or “every”.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the examples keep?** (A suspicion that the claim was true.)
- **What did the counterexample show?** (That the claim was false, for certain.)

WHAT MAKES THIS TRUSTWORTHY?

A counterexample is trustworthy because anyone can check it by doing the arithmetic. If 5×0.5 is 2.5, and 2.5 is less than 5, the claim is dead, whatever else is true.
A claim that survives all attempts to break it is still only "seen so far" until it has a reason.

PRACTICE

1. Claim: "All prime numbers are odd." Find a counterexample.
2. Claim: "The sum of two odd numbers is odd." Find a counterexample.
3. Claim: "Squaring a number makes it bigger than doubling it." Test n = 1, 2 and 3.
4. Fix the claim in problem 3 by saying which numbers it works for.
5. Kofi tried 30 numbers and the claim held. Does he have a proof?

BRIDGE *Bridge to the rest of the series:* In chapter 4, the Collatz game stayed in the "only seen so far" pile for everyone. Here claims move from "seen so far" to "refuted" by a single case. In Two Boxes, One Answer, one card where the boxes differ settles the question the same way.

PUZZLE PAGE

Spot the Slip

Three notebooks. One is perfect. The other two each hide a single slip. Find it!

TRACE

```
1 ( 37 × 43 )
2 redraw  →  ( ( 40 − 3 ) × ( 40 + 3 ) )
3 redraw  →  ( ( 40 × 40 ) − ( 3 × 3 ) )
4 pop ( 40 × 40 )  →  1,600
5 pop ( 3 × 3 )  →  9
6 pop ( 1,600 − 9 )  →  1,591
```

Notebook A ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 19 × 21 )
2 redraw  →  ( ( 20 − 1 ) × ( 20 + 1 ) )
3 redraw  →  ( ( 20 × 20 ) − ( 1 × ( 1 + 1 ) ) )
4 pop ( 20 × 20 )  →  400
5 pop ( 1 × ( 1 + 1 ) )  →  2
6 pop ( 400 − 2 )  →  398
```

Notebook B ☐ **perfect** ☐ **slip on line** _____

TRACE

```
1 ( 64 × 76 )
2 redraw  →  ( ( 70 − 6 ) × ( 70 + 6 ) )
3 redraw  →  ( ( 70 × 70 ) − ( 6 × 6 ) )
4 pop ( 70 × 70 )  →  4,900
5 pop ( 6 × 6 )  →  42
6 pop ( 4,900 − 42 )  →  4,858
```

Notebook C ☐ **perfect** ☐ **slip on line** _____



MULTIX // HINT

A slip can hide in a **pop** (wrong value) or in a **redraw** (value changed). Check each line against the problem at the top.

CHAPTER 7

The Four Parts of a Tidy Proof

STORY

By now Kofi has a messy notebook full of discoveries. When he wants to show a friend, he needs a tidy one. Kofi found that every good tidy proof has four parts. Here is a claim he wants to share:
Any three numbers in a row add up to three times the middle one.

PART 1: THE CLAIM

Say exactly what you are going to show. Kofi writes it as a plain sentence.

If you add three numbers in a row, the total is the middle number times 3.

The claim must be clear enough to test. “Three numbers in a row” means numbers such as 4, 5, 6, or 10, 11, 12.

PART 2: THE PLAN

Say, in a sentence, how you will show it. Kofi’s plan:

Give the middle number a name, then write the other two in terms of it.

The plan is not the proof. It tells the reader where to look.

Trust check. A reader can follow each step on their own, so they need not take your word.

PART 3: THE STEPS

Now do it, one small step at a time. Kofi names the middle number m. The number before it is m − 1 and the number after it is m + 1.

The −1 and the +1 cancel, which is why naming the middle number was a good choice in The Choice Before the Calculation.

TRACE

```
( ( m − 1 ) + m + ( m + 1 ) )
redraw  →  ( ( m + m + m ) + ( 1 − 1 ) )
pop ( 1 − 1 )  →  0
redraw  →  ( 3 × m )
```

Every step keeps the value, so the total equals 3 × m for any m.

PART 4: THE CHECK

Finish by testing the claim on a case, so a reader sees it work. Kofi picks two.

TRACE

```
4 + 5 + 6  =  15      and   3 × 5   =  15   ✓
10 + 11 + 12 = 33     and   3 × 11  =  33   ✓
```

A check does not prove the claim. The steps did that. The check protects against a slip in the steps, and it lets a reader see the claim with their own eyes.

STEP 5 What the tidy proof leaves out

The tidy proof left out the messy notebook: the first guesses, the dead ends, and the moment when Kofi noticed 5 was always in the middle of 4 and 6. As in chapter 3, the tidy notebook is for telling, and the messy notebook is for finding.

But a good tidy proof leaves a few things in:

- the **name** and what it stands for (m is the middle number),
- the **reason** for each step (every redraw keeps the value),
- the **check** at the end.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the four parts keep?** (The claim, the reason it is true, and a test of it.)
- **What did we drop?** (The order in which Kofi discovered it.)

WHAT MAKES THIS TRUSTWORTHY?

A reader can follow each step on their own, so they need not take your word. The plan tells them where you are going, and the check shows them it works.

PRACTICE

1. Use the four parts to show that the sum of two numbers in a row is odd. (Name the first number n.)
2. Check your claim in problem 1 with 7 + 8 and 20 + 21.
3. Write the claim and plan only for: “Any four numbers

in a row add to an even number." Do not do the steps.

4. Which part of a proof says what you are going to show?

5. Why is the check at the end not a proof?

BRIDGE *Bridge to the rest of the series:* In chapter 1, the tidy proof ended with a check. Here the four parts are named. In The Choice Before the Calculation, choosing the middle number as the name is what made the steps short.
Next in this book: Chapter 8: Three Routes, One Answer.

PUZZLE PAGE

Color by Answer

Shade every box whose answer is exactly **2,400**. Leave the others blank. What appears?

2403 + 3	75 × 32	2362 + 38	2381 + 24	2393 + 3	2402 − 2	2395 + 5	2399 + 6
48 × 50	2409 − 9	2438 − 38	2418 − 18	2415 − 15	2398 + 2	160 × 15	2398 + 2
2424 − 24	2406 − 6	6 × 400	125 × 125 − 115 × 115	40 × 60	2416 − 16	2364 + 36	2409 − 9
2376 + 24	2414 − 14	2376 + 24	2403 − 3	150 × 16	2375 + 25	2373 + 27	2386 + 14
4 × 597	49 × 49 − 1 × 1	2423 − 23	2434 − 34	75 × 32	2436 − 36	2436 − 36	2378 + 20
2380 + 21	2373 + 21	2433 − 33	2395 + 5	2415 − 15	8 × 300	2370 + 21	3 × 796
2396 + 5	2397 + 9	2389 + 2	2372 + 28	2413 − 13	2380 + 14	2399 + 11	2388 + 18
2367 + 27	2374 + 22	99 × 99 − 86 × 86	2379 + 14	2402 + 10	2374 + 27	2365 + 29	2396 + 8



MULTIX//SCAN

Some boxes look close and are not. Check the corner on each one.

CHAPTER 8

Three Routes, One Answer

STORY

Kofi had a map with a camp marked at the bottom of a hill. Three hikers started from the same place at the top, each on a different trail. The question for this chapter is simple: **do different trails reach the same camp?** In maths, the hill is a problem, the trails are different orders and methods, and the camp is the answer.

STEP 1 Three routes to 48 × 52

Here are three ways to reach the same number, from the case file.

TRACE

```
Route 1:  48 × 52  →  ( 50 × 50 ) − ( 2 × 2 )  =  2,500 − 4      =  2,496
Route 2:  48 × 52  →  ( 48 × 50 ) + ( 48 × 2 ) =  2,400 + 96     =  2,496
Route 3:  48 × 52  →  ( 50 × 52 ) − ( 2 × 52 ) =  2,600 − 104    =  2,496
```

The routes look nothing alike. Route 1 balances into a square. Route 2 splits 52 into 50 + 2. Route 3 splits 48 into 50 − 2. All three reach **2,496**.

STEP 2 Different orders of popping

Even within one route, there can be a choice about what to pop first. Take Route 1:

TRACE

```
( ( 50 × 50 ) − ( 2 × 2 ) )
```

Two circles are ready: (50 × 50) and (2 × 2). You can pop either one first.

TRACE

```
Order A:   pop ( 50 × 50 ) → 2,500,   pop ( 2 × 2 ) → 4,   pop ( 2,500 − 4 ) → 2,496
Order B:   pop ( 2 × 2 ) → 4,   pop ( 50 × 50 ) → 2,500,   pop ( 2,500 − 4 ) → 2,496
```

Two orders, one answer. The two small circles do not touch each other, so the order cannot change what happens.

Trust check. Agreement between routes is a good reason to trust an answer, because the routes shared no steps.

STEP 3 Counting routes and snapshots

Kofi tried counting for a small case:

TRACE

```
( ( 2 × 3 ) + ( 4 × 5 ) )
```

Call the circles A = (2 × 3), B = (4 × 5), and C, the big one. C cannot be popped until A and B are done.

- Orders: **A, B, C** or **B, A, C**. That is **2 orders**.
- Snapshots (what the notebook could look like at a moment): start; after A; after B; after A and B; after all. That is **5 snapshots**.

Both orders end with 6 + 20 = **26**.

Then he tried a bigger one with five circles:

TRACE

```
( ( ( 2 × 3 ) + ( 4 × 5 ) ) × ( 6 − 1 ) )
```

Call the circles A = (2 × 3), B = (4 × 5), D = (6 − 1), S for the sum, and P for the final product. S needs A and B first, and P needs S and D first. The valid orders are:

TRACE

```
A B D S P    A B S D P    A D B S P    B A D S P
B A S D P    B D A S P    D A B S P    D B A S P
```

There are **8 orders**. The snapshots are **11**. Every order ends at (26 × 5) = **130**.

Many orders, fewer snapshots, one answer. The orders weave together and pass through the same snapshots.

STEP 4 Two different things

Kofi noticed that there are two different questions here.

- **Agreement.** If two routes both finish, do they give the same answer? Yes, whenever each redraw keeps the value and each pop is done correctly.
- **Discovery.** Does a given route actually *finish*? Not every trail does. A rule like “add 4, then take away 4, then add 4 again” can go on forever and never reach the camp.

A trail that never arrives does not move the camp. The answer is still 2,496 even if one hiker walks in circles.

STEP 5 Using disagreement as a check

When two routes **disagree**, the camp has not moved. One of the routes has a slip, but the disagreement does not say which. You have to look at each step.

WHAT DID WE KEEP?

Write the answers in your notebook.

- **What did the different routes keep?** (The value, since every redraw preserved it.)
- **What did they change?** (The steps, the order, and the amount of work.)

WHAT MAKES THIS TRUSTWORTHY?

Agreement between routes is a good reason to trust an answer, because the routes shared no steps. It is stronger when the routes are very different from each other. It is not a proof that both are free of slips, but two independent routes would have to slip in the same way to agree on a wrong answer. Kofi marks an answer reached by two different routes as **has a reason**.

PRACTICE

1. Find **49 × 51** by two routes: around 50, and by 49 × 50 + 49 × 1. Do they agree?
2. For ((1 + 2) × (3 + 4)), name the circles, count the orders and the snapshots, and find the value.
3. A student finds 38 × 42 two ways. Route 1: 1,600 − 4. Route 2: 38 × 40 + 38 × 2 = 1,520 + 66. Do they agree? What do you do?
4. True or false: if two routes agree, the answer must be right. Explain.
5. Why does a hiker who never finishes not change the answer?

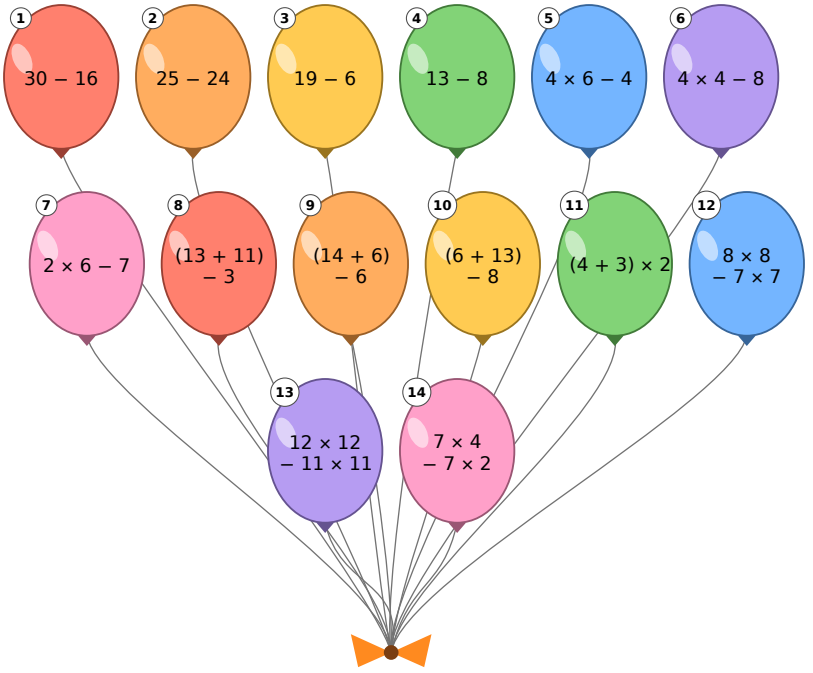
BRIDGE *Bridge to the rest of the series:* In The Boundary of a Step, one of the routes is lazy and one is eager, and the lazy one can arrive when the eager one cannot. In Two Boxes, One Answer, two boxes give the same answer at different cost.

PUZZLE PAGE

The Balloon Message

What is Multix's advice? Work out each balloon, use the key to turn the number into a letter, and fill the banner box with the same number.

A = 1	B = 2	C = 3	D = 4	E = 5	F = 6	G = 7	H = 8	I = 9	J = 10	K = 11	L = 12	M = 13
N = 14	O = 15	P = 16	Q = 17	R = 18	S = 19	T = 20	U = 21	V = 22	W = 23	X = 24	Y = 25	Z = 26



What was the answer?

													!
1	2	3	4	5	6	7	8	9	10	11	12	13	14

Answers (turn the book around)

Chapter 1

1. The middle is 15, and 15 times 15 is 225. We need 221, so the difference is 4, which means d is 2. The numbers are **13 and 17**. Check: 13 plus 17 is 30, and 13 times 17 is 221.
2. 100 minus 99 is 1, so d is 1. The numbers are **9 and 11**.
3. 100 minus 91 is 9, so d is 3. The numbers are **7 and 13**.
4. In the messy notebook, the products never get above 100. The best you can do is 10 and 10. In the tidy method, 100 minus d times d would have to equal 101, which would mean d times d is negative 1. No ordinary number multiplied by itself gives a negative answer. So no pair of ordinary numbers works. The messy notebook gives you a hunch, and the tidy proof tells you *why* the hunch is right.
5. (7 × 13) pops to 91, which is not 96. The sum (7 + 13) pops to 20, so the sum is fine, but the product is too small.

Color by Answer (1,225). The shaded boxes make a heart.



Chapter 2

1. 1 + 3 + 5 + 7 + 9 + 11 = **36**, which is 6 × 6.
2. A 6 by 6 square has 36 dots. The L-border for a 7 by 7 square has 6 + 6 + 1 = 13 dots, and 36 + 13 = **49**, which is 7 × 7.
3. (6 × 6) pops to 36, then 36 + 6 = 42, then 42 + 41 = **83**. The number 83 is prime.
4. A hundred examples is strong evidence that the pattern holds in the cases tried. It does not say what happens in the cases nobody has tried. Dani needs a reason, or has to say “only seen so far”.
5. (41 × 41) + 41 + 41 = 41 × 41 + 41 × 2 = 41 × 43, so the answer is **1,763**, and it can be divided by 41.
6. (a) has a reason, the L-border. (b) only seen so far, and in fact false.

Balloon Message.

1.	17 + 2	= 19	S
2.	6 − 5	= 1	A
3.	27 − 2	= 25	Y
4.	25 − 2	= 23	W
5.	3 × 5 − 7	= 8	H
6.	5 × 2 − 9	= 1	A
7.	5 × 5 − 5	= 20	T
8.	(15 + 13) − 3	= 25	Y
9.	(13 + 6) − 4	= 15	O
10.	(15 + 10) − 4	= 21	U
11.	(8 + 11) − 8	= 11	K
12.	2 × 10 − 2 × 3	= 14	N
13.	3 × 2 + 3 × 3	= 15	O
14.	12 × 12 − 11 × 11	= 23	W

Message: **SAY WHAT YOU KNOW!**

Answers (turn the book around)

Chapter 3

1. There are 20 numbers, so 10 pairs, each making 21. (10 × 21) pops to **210**.
2. 4 pairs, each making 9. (4 × 9) pops to **36**.
3. The first pair is 5 + 14 = 19. There are 10 numbers, so 5 pairs. (5 × 19) pops to **95**.
4. 49 pairs of 100, and 50 is left in the middle. (49 × 100) + 50 = 4,900 + 50 = **4,950**.
5. The messy one holds the check and the discovery. The check is a second road to the same answer, and it is worth keeping.

Spot the Slip.
Notebook A: slip on line **3**. The corner is 2 × 2, not 2 × (2 + 1). The answer should be 1,596.
Notebook B: slip on line **5**. 1 × 1 is 1. The answer should be 624.
Notebook C: **perfect**. 46 × 54 = 2,484.

Chapter 4

1. 5 → 16 → 8 → 4 → 2 → 1, which is **5 moves**.
2. 10 → 5 → 16 → 8 → 4 → 2 → 1, which is **6 moves**. It is the trail for 5 with one extra move at the start.
3. 12 → 6 → 3 → 10 → 5 → 16 → 8 → 4 → 2 → 1, which is **9 moves**.
4. 1 → 4 → 2 → 1. It goes around in a loop, so the game would never stop if it did not end at 1.
5. No. Twenty examples are evidence only. He should write **only seen so far**.

Color by Answer (899). The shaded boxes make a heart.



Answers (turn the book around)

Chapter 5

1. 15 = 14 + 1 and 21 = 20 + 1. The pieces are 14 × 20 = 280, 14 × 1 = 14, 1 × 20 = 20, and 1 × 1 = 1. Their sum is 280 + 14 + 20 + 1 = 315, which is **odd**.
2. **Odd.** Both numbers are odd, and odd times odd is odd by the four-piece reason.
3. 14 is even and 9 is 8 + 1. So the sum is even + even + 1, which is **odd**.
4. **Even.** 8 is even, so 8 × 25 is eight lots of 25, which pair up with none left over. (It equals 200.)
5. Each even number can be shared into pairs with none left over. Putting two such numbers together still leaves none over, so the sum is even.

Balloon Message.

1.	5 − 3	= 2	B
2.	10 − 5	= 5	E
3.	27 − 16	= 11	K
4.	8 × 2 − 7	= 9	I
5.	2 × 8 − 2	= 14	N
6.	6 × 2 − 8	= 4	D
7.	4 × (12 − 7)	= 20	T
8.	(13 + 5) − 3	= 15	O
9.	(15 + 15) − 4	= 26	Z
10.	3 × 3 − 2 × 2	= 5	E
11.	3 × 11 − 3 × 5	= 18	R
12.	4 × 4 − 1 × 1	= 15	O

Message: **BE KIND TO ZERO!**

Chapter 6

1. The number **2** is prime and even.
2. For example 3 + 5 = 8, which is even. (Any two odd numbers will do.)
3. n = 1: 1 against 2, so squaring is smaller (counterexample). n = 2: 4 and 4, the same. n = 3: 9 against 6, bigger. The claim fails at 1 and 2.
4. “Squaring a number is bigger than doubling it when the number is bigger than 2.” For n = 3, 4, 5: 9 > 6, 16 > 8, 25 > 10.
5. No. Thirty examples are evidence only, and one counterexample somewhere could still break it.

Spot the Slip.
Notebook A: **perfect**. 37 × 43 = 1,591.
Notebook B: slip on line **3**. The corner is 1 × 1, not 1 × (1 + 1). The answer should be 399.
Notebook C: slip on line **5**. 6 × 6 is 36. The answer should be 4,864.

Answers (turn the book around)

Chapter 7

1. The numbers are n and n + 1. Their sum is (n + n + 1), which is ((2 × n) + 1). Two lots of n is even, and adding 1 makes it odd.
2. 7 + 8 = 15, which is odd. ✓ 20 + 21 = 41, which is odd. ✓
3. Claim: “The sum of any four numbers in a row is even.” Plan: “Name the first number n, write the four numbers in terms of n, and show the total is 2 times something.”
4. The **claim**.
5. A check tests particular cases. The claim covers every case, and only the steps cover those.

Color by Answer (2,400). The shaded boxes make a heart.



Chapter 8

1. Route 1: 2,500 − 1 = 2,499. Route 2: 2,450 + 49 = 2,499. They **agree**.
2. Circles: (1 + 2), (3 + 4), and the product. Orders: **2**. Snapshots: **5**. Value: 3 × 7 = **21**.
3. Route 1 gives 1,596 and Route 2 gives 1,586, so they **disagree**. There is a slip somewhere. Check each step: 38 × 2 is 76, not 66. The correct sum is 1,520 + 76 = **1,596**.
4. **False**, strictly. It is a good reason, but two routes could slip in the same way. It is far more trustworthy than one route alone.
5. The answer belongs to the problem, not to the route. A route that does not arrive has just not found it.

Balloon Message.

1.	30 − 16	= 14	N
2.	25 − 24	= 1	A
3.	19 − 6	= 13	M
4.	13 − 8	= 5	E
5.	4 × 6 − 4	= 20	T
6.	4 × 4 − 8	= 8	H
7.	2 × 6 − 7	= 5	E
8.	(13 + 11) − 3	= 21	U
9.	(14 + 6) − 6	= 14	N
10.	(6 + 13) − 8	= 11	K
11.	(4 + 3) × 2	= 14	N
12.	8 × 8 − 7 × 7	= 15	O
13.	12 × 12 − 11 × 11	= 23	W
14.	7 × 4 − 7 × 2	= 14	N

Message: **NAME THE UNKNOWN!**